EE3404: Introductory Electric Circuits

Paired Quizzes, Classroom Questions and Individual Assignment

Second-Year Bachelor of Electronics Engineering
National University of Lesotho

Coverage: basic electrical quantities; charge and current; voltage; power and energy; passive sign convention; active and passive elements; sources; resistors; Ohm's law; open and short circuits; series and parallel connections; Kirchhoff's Current Law (KCL); and Kirchhoff's Voltage Law (KVL).

A. Short Paired Quizzes

10 short quizzes: approximately 5–8 minutes each.

1. Charge and Current

A conductor carries a current $i(t) = 2t + 1$ A for $0 \le t \le 3$ s. Calculate the total charge transferred during the 3 s interval. Then explain physically what the result means.

2. Voltage, Power and Energy

A 12 V source supplies a constant current of 2.5 A to a device for 4 minutes. Determine: (a) the power delivered; and (b) the energy transferred in joules and Wh. Discuss why power and energy should not be used interchangeably.

3. Passive Sign Convention

An element has $v = 10$ V across it and $i = 3$ A. The current enters the terminal marked positive. Determine the power and state whether the element is absorbing or supplying power. What changes if the current direction is reversed?

4. Active or Passive?

Classify a battery, resistor, capacitor and inductor as active or passive elements. Then discuss why the statement "a passive element cannot contain energy" is incorrect.

5. Ohm's Law and Power

A 1.2 kΩ resistor is connected across 12 V. Calculate its current and power dissipation. If the voltage doubles while R remains constant, by what factor does the power change? Explain mathematically.

6. Open versus Short Circuit

Compare an ideal open circuit and an ideal short circuit in terms of R, I and V. A student claims: "A short circuit has zero current because its voltage is zero." Is the student correct? Explain using Ohm's law and physical reasoning.

7. Series or Parallel?

Two resistors carry the same current. Does this necessarily mean that they are connected in series? Two resistors have equal voltages across them. Does this necessarily mean that they are in parallel? Defend both answers using the definitions of series and parallel connections.

8. KCL Challenge

At a node, currents of 5 A and 2 A enter while 4 A leaves. Determine the magnitude and direction of the fourth current. Explain how your result illustrates conservation of charge.

9. KVL and Conservation of Energy

A loop contains a 24 V source and three voltage drops of 5 V, 8 V and Vx. Determine Vx. Explain why a value that does not satisfy KVL would imply a violation of conservation of energy.

10. Real-World Scenario: USB/Phone Charging

A 5 V USB supply delivers a constant 0.50 A to a phone for 3 h. Calculate the charge delivered in coulombs and the electrical energy supplied in Wh and joules. If only 80% reaches the battery, determine the energy stored. Discuss where the remaining energy might go.

B. Paired Classroom Problem-Solving Questions

5 questions: approximately 10–15 minutes each.

1. From Charge to Power

The charge passing a point in a circuit is $q(t) = 2t^3 - 3t^2 + 4t$ C. For t = 2 s:

(a) Determine the instantaneous current.

(b) If the voltage across the element is 8 V, determine its instantaneous power.

(c) State whether it absorbs or supplies power if the current enters its positive terminal.

(d) Explain the physical connection between charge → current → power → energy.

2. Resistor Selection

An electronic subsystem requires approximately 20 mA from a 12 V DC supply.

(a) Calculate the required resistance.

(b) Determine the resistor power dissipation.

(c) Would you recommend a 0.25 W resistor? Justify quantitatively.

(d) If the resistance changes by +10%, calculate the new current and discuss the engineering consequence.

3. KCL as a Diagnostic Tool

At an electronic circuit node, measured currents are I_1 = 12 mA entering, I_2 = 7 mA leaving, and I_3 = 3 mA leaving.

(a) Use KCL to determine I_4.

(b) State its direction.

(c) A technician instead measures I_4 = 4.5 mA. Does the measurement set satisfy KCL?

(d) Discuss at least two engineering reasons why experimental measurements might not balance exactly even though KCL remains valid.

4. Circuit Topology

A student says: "If two resistors have the same current, they are in series; if they have the same voltage, they are in parallel."

Critically assess this statement. Sketch one circuit that supports the student's reasoning and one counterexample showing why circuit connectivity, rather than coincidentally equal measured values, determines series/parallel relationships.

5. Real-World Scenario: Lesotho Solar-Powered Electronics

A small solar-powered electronics unit operates from a regulated 12 V DC supply. Its controller draws 0.15 A and two identical sensors each draw 50 mA. Assume all three loads are connected across the 12 V supply.

(a) Draw an appropriate circuit representation.

(b) Use KCL to determine the source current.

(c) Calculate the power consumed by each load and the total load power.

(d) Determine the energy required for 8 h of operation.

(e) If only 85% of energy drawn from the battery reaches the loads, determine the minimum battery energy that must be supplied.

(f) Explain why the loads should be connected in parallel rather than series.

C. Individual Assignment

5 challenging questions requiring full working, interpretation and engineering reasoning.

Question 1: Time-Varying Current, Charge and Energy

The current entering the positive terminal of an electronic device is $i(t) = 4e^{-2t}$ A for $t \geq 0$, and its voltage is a constant 10 V.

(a) Determine the charge transferred from t = 0 to t = 2 s.

(b) Obtain an expression for instantaneous power p(t).

(c) Calculate the energy absorbed over the same interval.

(d) Determine the total charge and energy that would ultimately be transferred as $t \to \infty$.

(e) Explain the physical significance of the current approaching zero while total transferred charge approaches a finite value.

Question 2: Ohm's Law, Power and Design Judgement

An LED-based electronic unit is represented, for this exercise, by a 180 Ω resistive load connected to a 12 V source.

(a) Determine its current and power consumption.

(b) Calculate the energy consumed during 10 h of continuous operation.

(c) Repeat part (a) if the supply voltage rises by 10%.

(d) Calculate the percentage increase in power.

(e) Explain why a relatively small increase in voltage can produce a substantially larger increase in resistor heating.

(f) Recommend an appropriate minimum resistor power rating, giving an engineering justification rather than merely selecting the calculated power.

Question 3: KCL, KVL and Power Conservation

A 24 V DC source supplies two parallel branches. Branch 1 contains a 12 Ω resistor and Branch 2 contains a 24 Ω resistor.

(a) Draw and label the circuit.

(b) Determine each branch current using Ohm's law.

(c) Verify KCL at the source node.

(d) Apply KVL independently to each branch.

(e) Calculate the power supplied by the source and the power absorbed by each resistor.

(f) Demonstrate quantitatively that total power supplied equals total power absorbed.

(g) Explain why the simultaneous satisfaction of KCL, KVL and power conservation provides a useful check on circuit-analysis calculations.

Question 4: Fault Analysis: Open and Short Circuits

A 12 V supply is connected to a 120 Ω load.

(a) Determine the normal load current and power.

(b) The load subsequently fails open circuit. Determine the ideal current and explain what happens to the voltage across the open terminals.

(c) The load instead fails short circuit. Explain what the ideal circuit model predicts for current and why this result is physically problematic.

(d) If the real source has an internal resistance of 0.50 Ω, calculate the short-circuit current and the power dissipated internally.

(e) Compare the safety consequences of open- and short-circuit faults and explain why practical power supplies require protection.

Question 5: Real-World Engineering Design: Remote Monitoring Station

A remote environmental-monitoring station in Lesotho is powered from a 12 V battery. Its electronics consist of a controller R_1 = 120 Ω, communication module R_2 = 60 Ω, and sensor unit R_3 = 240 Ω. The three loads operate simultaneously and are connected in parallel.

(a) Draw the circuit with correct voltage polarities and current directions.

(b) Calculate the current through each load.

(c) Apply KCL to determine the total battery current.

(d) Determine the equivalent resistance seen by the battery.

(e) Calculate the power absorbed by each component and the total power supplied by the battery.

(f) Verify conservation of power.

(g) If the station operates continuously for 24 h, determine its daily energy requirement in Wh.

(h) If the battery has a usable capacity of 20 Ah at 12 V, estimate the ideal operating time.

(i) One sensor branch becomes open-circuited. Recalculate the total current and power and explain what happens to the other parallel loads.

(j) The communication branch instead develops an ideal short circuit. Explain, using circuit laws, why this fault is fundamentally more dangerous than the open-circuit fault and identify the practical limitation of the ideal calculation.

Instruction to students: Show all working, state assumptions clearly, use correct SI units, and support qualitative conclusions with appropriate circuit laws and engineering reasoning.